

Blibli.com: Analyzing customer ecommerce behavior with BigQuery

Tell us your challenge. We're here to help.

[Contact us \(/contact\)](#)

[/about-blibli.com/](#)

Indonesian ecom
business Blibli.cc
turned to Google
scale to support
growth, give its a
team faster acce
and democratize
access.



About Blibli.com

Founded in 2010 and launched in 2011,
Blibli.com is an Indonesian business-to-
consumer ecommerce provider.

Industries: Retail & Consumer Goods

Location: Indonesia

[\(https://cloud.google.com/compute/\)](https://cloud.google.com/compute/)

Google Cloud Results

- Extends access to data warehousing and analysis to all relevant team members
- Delivers the seamless, cost-effective scalability needed to support business growth
- Enables the IT team to shift focus from infrastructure tasks to helping the business obtain value from data

Cut time to provision analytics by almost 97%

BigQuery (<https://cloud.google.com/bigquery/>)

Cloud Pub/Sub (<https://cloud.google.com/pubsub/>)

Dataflow (<https://cloud.google.com/dataflow/>)

Cloud Dataproc
(<https://cloud.google.com/dataproc/>)

Compute Engine
(<https://cloud.google.com/compute/>)

With more than 260 million people, Indonesia represents a considerable opportunity for business-

to-consumer ecommerce companies. These companies can sell products and services to consumers and act as marketplaces for other vendors to sell to consumers.

Blibli.com (<https://www.blibli.com/>) aims to become a leader in Indonesia's ecommerce sector. Founded in 2010 and launched in 2011, Blibli.com plans to be a one-stop shop for consumers seeking products and services in mobile and tablet devices, sports and outdoor activities, home and living, men's and women's fashion, tickets and vouchers, and other categories. The business also provides bill payment and hotel and airline booking services.

Over the past seven years, Blibli.com has grown to employ more than 1,200 people and has offices and facilities in eight cities in Indonesia.

“The simplicity
and scalability of
Google Cloud
was ideal for our
needs.”

—Welly Dwi Putra,
Principal Data Engineer,
Blibli.com

Blibli.com relies on an ecommerce engine and a data platform to run its business. The data platform captures clickstream data as well as data from its backend systems. The business uses the clickstream data to capture intelligence about customer behavior on its website, including trending products and services, while backend system data is used to measure performance against business metrics and key performance indicators.

Growth exposes limitations

Blibli.com started running its ecommerce and data platform on infrastructure in an on-premises data center. However, the speed of Blibli.com's growth soon exposed the inability of this infrastructure to scale quickly and cost effectively. The Blibli.com IT team needed four hours or longer to provide ready-to-consume data from its data warehouse to its team of analysts. In addition, the IT team had to control and limit access to its data platform, as a CPU-intensive query could overwhelm its servers and compromise performance.

Finally, the IT team was spending too much time managing its infrastructure rather than helping the business extract value from its data.

The IT team began exploring alternative infrastructure options for Blibli.com's data platform. However, many offerings required team members to manually complete tasks, such as RAM and CPU allocation to virtual machines and partitioning to optimize query performance.

The team also established through its market review that machine learning presented considerable opportunities to improve its systems and use of data.

“With BigQuery
we can provide
access to
anyone who
needs it and not
be at all
concerned about
performance.
And because
Google Cloud
runs on Google
infrastructure,
we can be
confident our
data will be
available as we
need it.”

—Welly Dwi Putra,
Principal Data Engineer,
Blibli.com

A comprehensive evaluation of Google Cloud

The business then conducted a thorough evaluation of Google Cloud (<https://cloud.google.com/>) – in particular the BigQuery

(<https://cloud.google.com/bigquery/>) analytics data warehouse – and found the service aligned perfectly with its requirements. “The simplicity and scalability of Google Cloud was ideal for our needs,” says Welly Dwi Putra, Principal Data Engineer at Blibli.com. The business conducted a proof of concept and started moving its data platform to Google Cloud two months later, before completing its migration just four months after that.

Blibli.com is now running its data platform in a Google Cloud environment designed to optimize performance and availability. An open-source stream processing platform forwards clickstream and microservices data from its on-premises infrastructure to the Cloud Pub/Sub

(<https://cloud.google.com/pubsub/>) messaging service. Cloud Pub/Sub forwards the messages to BigQuery for analysis and visualization. The business uses Cloud Dataflow (<https://cloud.google.com/dataflow/>) to transform and enrich approximately 23GB per day of streaming data and 50GB per day of batch data. All data is stored on Google Cloud for archival, disaster recovery, and ease of access.

Compute Engine (<https://cloud.google.com/compute/>)

provides infrastructure as a service that enables Bibli.com's IT team to easily create virtual machine images and size clusters for systems such as an Elasticsearch search engine, which gives the business the ability to search pre-aggregated data used by its internal services.

Bibli.com's analysis team uses the data in BigQuery to create dashboards or KPIs, while a third-party tool visualizes data to help managers and executives make sound business decisions. Cloud Dataproc (<https://cloud.google.com/dataproc/>) undertakes data processing that enables the business to suggest products or product combinations to website visitors, based on viewing histories.

“Machine
learning services
could improve
our product
combinations as
well as help us
automate and
streamline
inventory
planning,
customer

segmentation,
and events
classification.
We now have
more time to
consider our
options in this
area because
Google Cloud
has allowed us
to spend less
time managing
our
infrastructure.”

—Welly Dwi Putra,
Principal Data Engineer,
Bibli.com

Accelerating data availability

Bibli.com has achieved considerable benefits from its move to Google Cloud. The speed and scalability of BigQuery has cut the time to provide ready-for-consumption data to its analysis team from 4 hours to just 20 minutes. In addition, the business has been able to provide data access to any team member that requires it.

“With BigQuery we can provide access to anyone who needs it and not be at all concerned about performance,” says Dwi Putra. “And because Google Cloud runs on Google infrastructure, we can be confident our data will be available as we need it.

Machine Learning presents opportunities

Blibli.com is now well positioned to use services such as Cloud Machine Learning Engine (<https://cloud.google.com/ml-engine/>) to provide more relevant customer experiences and operate more efficiently. “Machine learning services could improve our product combinations as well as help us automate and streamline inventory planning, customer segmentation, and events classification,” says Dwi Putra. “We now have more time to consider our options in this area because Google Cloud has allowed us to spend less time managing our infrastructure.”

With Google Cloud, Blibli.com is now well positioned to realize its goal of becoming a one-stop shop for online consumers in Indonesia. The business has a robust foundation to achieve ambitious business growth targets over the coming years.